University of Namibia Computer

programming 1

SRS FILE:

Group 8: Triad of Terror

ENGITRIAD

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System Requirements Specification

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Section 1 — System Overview

1.1 System Description

EngiTriad is a multi-domain mobile application built with Expo, React Native, and Firebase, combining three specialised engineering tools into a single platform.

- The Corrosion Estimation module allows metallurgical and civil engineers to input material type, environmental conditions, and exposure duration to obtain a calculated corrosion rate, saved to Firestore per user.
- The Concrete Mixer Calculator accepts project inputs such as concrete grade and required volume, returning precise quantities of cement, sand, aggregate, and water — with each calculation stored for future retrieval.
- The Blasting Planner supports mining engineers in scheduling and documenting blasting events, including materials, crew assignments, and exclusion zones, with real-time Firestore synchronisation giving all authorized personnel live visibility of operations.

All three modules sit behind a shared Firebase Authentication layer, ensuring every record is tied to a verified user account.

1.2 Target Users

EngiTriad serves three primary user groups across its modules:

User Profile	Domain	Primary Module	Tech Level
Metallurgical / Civil Engineer	Metallurgical & Civil	Corrosion Estimation	Intermediate
Civil Site Engineer / Supervisor	Civil Engineering	Concrete Mixer Calculator	Intermediate
Mine Supervisor / Blasting Officer	Mining Engineering	Blasting Planner	Moderate
Engineering Students (UNAM)	All three domains	All modules	Beginner–Intermediate

1.3 Technology Overview

Technology	Role in EngiTriad
Expo (Managed Workflow)	Development platform; Expo Go for testing during development, EAS Build for final Android APK.
React Native + React Navigation	All screens, forms, and a bottom tab navigator (one tab per module) built in JavaScript/JSX.
Firebase Authentication	Email/password registration and login; all Firestore data tied to authenticated user UIDs.
Cloud Firestore	Stores corrosion Records, concrete Mixes, blasting Events, and users' collections with auth-gated security rules.
Firebase Storage	Stores site photos or documents attached to blasting event records.

1.4 Assumptions and Constraints

- An active internet connection is required; offline data entry is not supported in this release.
- The primary target platform is Android 10.0 (API Level 29) and above; an Android APK is the required submission deliverable.
- Expo Managed Workflow uses the Firebase JavaScript SDK — @react-native-firebase is not used.
- Corrosion and concrete calculations use standard empirical formulas for indicative purposes and are not certified for structural safety sign-off.
- Firebase configuration keys must be stored in a .env file excluded from the public GitHub repository via. gitignore.

Section 2 — Firebase Backend Architecture

2.1 Overview

Firebase is used as the Backend-as-a-Service (BaaS) for this application. It handles authentication, realtime data storage, and optional file storage. The system relies entirely on Firebase services to ensure scalability, security, and real-time synchronisation across users.

2.2 Firebase Services Used

Service	Purpose	Status
Firebase Authentication	User registration, login, logout	Mandatory
Cloud Firestore	Store and retrieve application data in real-time	Mandatory
Firebase Storage	Store images and files	Optional
Firebase Cloud Messaging	Push notifications	Optional

2.3 Firebase Authentication

Firebase Authentication provides secure user identity management without requiring a custom backend. Key capabilities include:

- Email and password registration
- Secure login and logout
- Session persistence
- Unique User ID (UID) generation for each authenticated user
- Only authenticated users can access system features

2.3.1 Authentication Workflow

1. User enters login or sign-up details
2. Firebase verifies the credentials
3. If successful, the user is authenticated
4. A unique User ID (UID) is assigned
5. The application uses this UID to manage user-specific data

2.4 Cloud Firestore Database Design

Four Firestore collections are used, all secured with request.auth != null rules:

2.4.1 Collection: users

Field	Description
uid	Firebase-assigned user identifier
displayName	User's full name

email	Registered email address
role	engineer / supervisor / student
domain	Engineering domain of the user
createdAt	Timestamp of account creation

2.4.2 Collection: Corrosion Records

Field	Description
User Id	Linked user UID
Material Type	Type of material assessed
Environment Condition	Environmental exposure conditions
Exposure Duration	Duration of exposure
Corrosion Rate Result	Calculated corrosion rate
unit	Unit of measurement
notes	Optional notes
Created At	Record creation timestamp

2.4.3 Collection: Concrete Mixes

Field	Description
User Id	Linked user UID
Project Name	Name of the construction project
Concrete Grade	Grade/strength class of concrete
Volume Required	Total volume required (m^3)
Cement Qty	Calculated cement quantity
Sand Qty	Calculated sand quantity
Aggregate Qty	Calculated aggregate quantity
Water Qty	Calculated water quantity
Created At	Record creation timestamp

2.4.4 Collection: Blasting Events

Field	Description
User Id	Linked user UID
Event Name	Name/identifier of the blast event

Scheduled Date	Date and time of scheduled blast
Blast Location	Geographic/site location
Materials Required	List of materials needed
Crew Assignments	Personnel assignments
Exclusion Zone Radius	Safety exclusion zone radius (m)
status	Current event status
Attachment URL	URL of uploaded site file/photo
Created At	Record creation timestamp

2.5 Data Operations

Operation	Description
Create	Users create new records (calculations, events, profiles)
Read	Users retrieve data in real-time via Firestore listeners
Operation	Description
Update	Users edit existing records where permitted
Delete	Users remove records if they have the required permissions

2.6 Firebase Storage

Firebase Storage is used to store images and documents attached to blasting event records. File URLs are saved in the corresponding Firestore document upon successful upload, enabling seamless retrieval.

2.7 Security Rules

All Firestore collections and Storage buckets are protected by the following core rule: only authenticated users (request. auth! = null) can read or write data. This ensures no unauthorized access to application data.

2.8 Firebase Configuration & Environment Variables

Firebase is initialized using a configuration object stored securely within the project. All sensitive keys (API key, project ID, app ID, etc.) are stored in a .env file and excluded from the public GitHub repository via. gitignore, ensuring credentials are not exposed.

2.9 Integration with Expo

Firebase uses the JavaScript SDK and integrates with the Expo Managed Workflow without requiring any native module setup. This ensures compatibility with Expo Go during development and EAS Build for the final Android APK deliverable.

Section 3 — User Interface Design (Figma)

3.1 Overview

This section presents the user interface designs for the ENGITRIAD application. The designs were created using Figma and represent the visual layout and user experience of the application across all three engineering modules.

3.2 Design Tools & Specifications

Property	Value
Tool Used	Figma
Platform	Android Mobile
Design Type	High-fidelity wireframes and mockups

3.3 Colour Theme & Style Guide

Role	Hex Code	Usage
Primary Colour	#DDA131 (Amber)	Buttons, highlights, accent elements
Secondary Colour	#02153A (Navy Blue)	Headers, text, navigation bar
Background	#FFFFFF (White)	Screen backgrounds
Text Colour	#02153A (Navy Blue)	Body text and labels
Font Family	Inter	All UI text

3.4 Screen Designs

The following screens have been designed and are included in the Figma prototype:

3.4.1 Authentication Screens

- Splash Screen — displays ENGITRIAD logo and tagline
- Login Screen — email and password input fields, login button, and link to Sign Up
- Sign Up Screen — full name, email, password, and confirm password fields with Sign Up button • Forgot Password Screen — email input and Send Reset Link button

3.4.2 Main Navigation Screens

- Welcome Screen — welcome message and navigation to Departments
- Departments Screen — entry points to Cement Mixer, Corrosion, and Blast modules

3.4.3 Module Screens

- Concrete Slab Calculator — input fields for slab dimensions and a Calculate button

- Concrete Results Screen — displays calculated quantities of cement, sand, aggregate, and water
- Corrosion Rate Estimator — metal type, pH, and temperature inputs with an Estimate button
- Corrosion Results Screen — displays the calculated corrosion rate and unit
- Slope Stability Blast Calculator — slope angle, height, and cohesion inputs with a Calculate button
- Blast Results Screen — displays Factor of Safety and Stability Status

3.5 Navigation Flow

The application follows a linear authentication flow before branching into the three module tabs:

```
                    Splash Screen
                          ↓
          Login Screen ←→ Sign Up Screen
                          ↓
                   Welcome Screen
                          ↓
                 Departments Screen
        ↓                 ↓                 ↓
   Cement Mixer      Corrosion         Blast Planner
        ↓                 ↓                 ↓
     Results           Results           Results
```

Section 4 — Core Features

4.1 Module 1 — Corrosion Rate Estimator

Allows metallurgical and civil engineers to input material type, environmental conditions, and exposure duration to obtain a calculated corrosion rate. Results are persisted to the corrosion records Firestore collection under the authenticated user's account.

- Inputs: material type, environmental condition, exposure duration (years), optional notes
- Output: corrosion rate with unit, record saved to Firestore, history viewable per session

4.2 Module 2 — Concrete Mixer Calculator

Help civil engineers determine precise material quantities for a given concrete mix design. Results are stored in the concrete Mixes Firestore collection.

- Inputs: project name, concrete grade (e.g., C20/C25/C30), required volume (m^3)
- Outputs: cement, sand, aggregate, and water quantities; record saved to Firestore

4.3 Module 3 — Blasting Planner

Enables mine supervisors and blasting officers to plan, document, and track blasting events in real-time. Data synchronises across all authorised personnel via Firestore listeners.

- Inputs: event name, scheduled date, blast location, materials, crew assignments, exclusion zone radius, optional attachment
- Outputs: event record saved to blasting events; real-time sync for all users; status tracking (planned / in-progress / completed)

4.4 Shared Features

Feature	Description
Firebase Authentication	All modules require a verified user account; records are tied to authenticated UIDs
History & Retrieval	All calculation results and events are retrievable per authenticated user session
Real-Time Updates	Firestore listeners push data updates without requiring a manual page refresh
Secure Data Isolation	Each user’s records are scoped to their UID; no cross-user data access is possible
User Profile Management	Role, domain, and display name stored in the users Firestore collection

Section 5 — Non-Functional Requirements

The following non-functional requirements govern the quality attributes of EngiTriad:

Category	Requirement
Performance	Initial data loads within 3 seconds on 4G; calculation results display within 1 second; Firestore updates reflect within 2 seconds for all connected users.
Security	All Firestore reads/writes gated by request. auth! = null; Firebase keys stored in env and excluded from version control; passwords never stored in plaintext; data scoped per UID.
Usability	Clear labels, validation messages, and placeholder text on all forms; consistent bottom-tab navigation; Inter font and Amber/Navy colour theme applied throughout.
Reliability	Active internet connection required; Firebase availability is a system dependency; all users must register before accessing features; offline mode is out of scope.
Portability	Targets Android 10.0 (API Level 29) +; distributed as an Android APK via EAS Build; Expo Managed Workflow eliminates native module configuration requirements.